

**Department of Computer Technology****Vision Statement of the Department.**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)



Nagar Yuwak Shikshan Sanstha's

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Session	2026-27	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	13	Name of Student	Juhi Parwani

Practical Number	1(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	Develop a JavaScript program to model a shopping cart using an array of product objects. Each product should contain id, name, price, and quantity. The program should perform operations such as: • Adding products to the cart • Removing products using <code>filter()</code> • Extracting product information using <code>map()</code> • Calculating total cart price using <code>reduce()</code> • Using arrays, objects, arrow functions, and higher-order functions.
Problem Definition	Develop a JavaScript program to model a shopping cart using an array of product objects. Each product should contain ID, name, price, and quantity. The program should allow products to be added and removed from the cart, extract product information, and calculate the total cart price using JavaScript array methods such as <code>push()</code>, <code>filter()</code>, <code>map()</code>, and <code>reduce()</code>. The program also demonstrates the use of objects, arrays, arrow functions, and higher-order functions.
Theory (100 words)	JavaScript provides several features for storing and processing data efficiently. An array is a collection of multiple values stored in a single variable. An object stores data in the form of key-value pairs such as product ID, name, price, and quantity. An array of objects can be used to represent multiple products in a shopping cart.



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Procedure and Execution

(100 Words)

Step for Implementation:

Create the JavaScript file.

Create the shopping cart using an array of objects.

Implement the Add operation.

Implement product-name extraction using `map()`.

Calculate the total price using `reduce()`.

Add a new product using `push()`.

Remove a product using `filter()`.

Display product names and total price.

Execute the program.

Code:

```
!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Shopping Cart</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1>Shopping Cart</h1>

  <section id="product-list"></section>

  <aside id="cart">
    <h2>Your Cart</h2>
    <ul id="cart-list"></ul>
```



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```
<p>Total: ₹<span id="total-price">0</span></p>
</aside>

<script src="script.js"></script>

</body>
</html>

const products = [
  { id: 1, name: "Notebook", price: 40 },
  { id: 2, name: "Pen", price: 10 },
  { id: 3, name: "Bag", price: 500 },
  { id: 4, name: "WaterBottle", price: 150 }
];

let cart = [];

const productList =
document.getElementById("product-list");
const cartList = document.getElementById("cart-list");
const totalPriceEl =
document.getElementById("total-price");

function displayProducts() {
  productList.innerHTML = products
    .map((product) => `
      <div class="product-card">
        ${product.name} - ₹${product.price}
        <button
onclick="addToCart(${product.id})">
          Add to Cart
        </button>
      </div>
    `)
    .join("");
}

function addToCart(id) {
```



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```
const item = products.find((p) => p.id === id);
cart.push(item);
displayCart();
}

function removeFromCart(id) {
  cart = cart.filter((item) => item.id !== id);
  displayCart();
}

function calculateTotal() {
  return cart.reduce(
    (sum, item) => sum + item.price,
    0
  );
}

function displayCart() {
  cartList.innerHTML = cart
    .map((item) => `
      <li>
        ${item.name} - ₹${item.price}
        <button
          onclick="removeFromCart(${item.id})">
          ✕
        </button>
      </li>
    `)
    .join("");

  totalPriceEl.textContent = calculateTotal();
}

displayProducts();
body {
  font-family: Arial, sans-serif;
  display: flex;
```



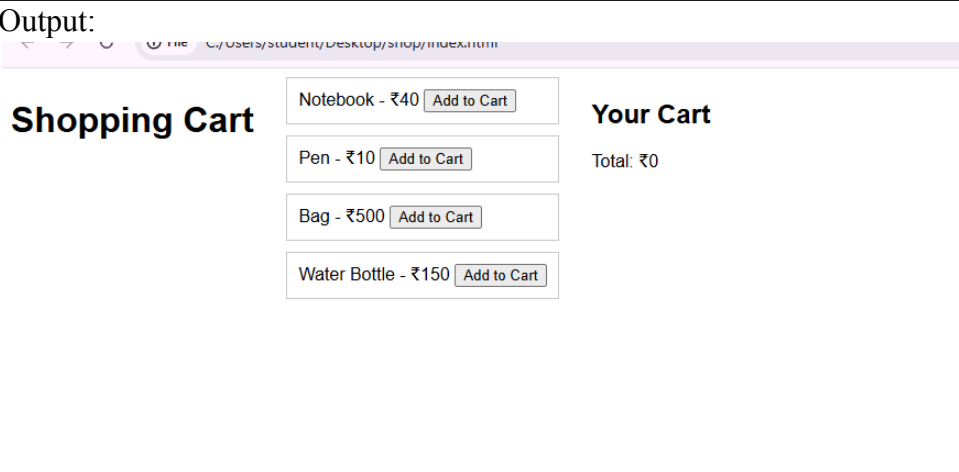
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	<pre>gap: 30px; } .product-card { border: 1px solid #ccc; padding: 10px; margin-bottom: 10px; }</pre> Output: 
Output Analysis	The output displays a Shopping Cart Web Application with a list of available products and an Add to Cart button for each product. When a product is added, it appears in the Your Cart section. The user can remove an item using the ✖ button, and the cart total is automatically updated. The program uses map() to display products and cart items, filter() to remove items, find() to locate a selected product, push() to add products to the cart, and reduce() to calculate the total price. Thus, the application demonstrates dynamic DOM manipulation and the use of JavaScript higher-order functions.
Link of student Github profile where lab assignment has been uploaded	https://github.com/juhiparwa-1507/Web-Technology-Lab.git
Conclusion	The JavaScript program successfully models a shopping cart using an array of objects and demonstrates push(), map(), filter(), and



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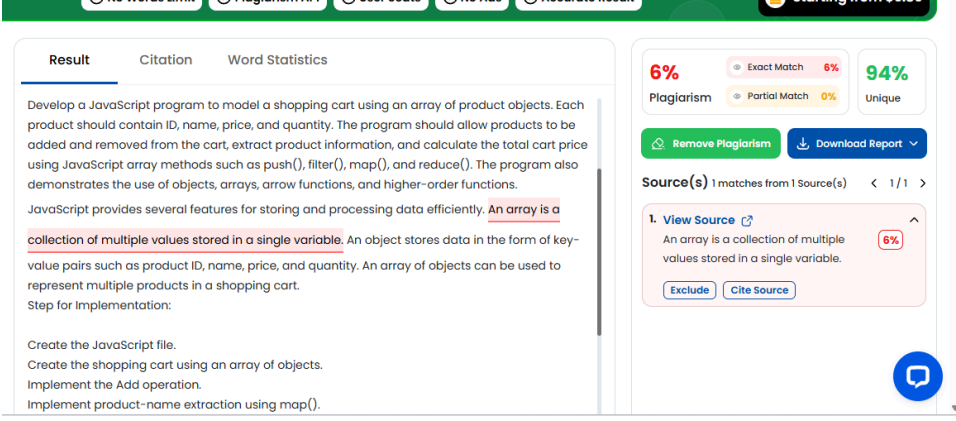
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	<code>reduce()</code> for adding products, removing products, retrieving product names, and calculating the total price. .
Plag Report (Similarity index < 12%)	Plagiarism Report: To be attached. 
Date	05-10-26